

THERMAL MODELLING, DESIGN AND EVALUATION OF A CRYOGENIC COOLING SYSTEM FOR A BEAMLINE ENDSTATION

D. Tillin[†], S. Beamish, C. Bovo, D. Burn, S. Davies, J. H. Kelly

Diamond Light Source, Harwell Science and Innovation Campus, Didcot, Oxfordshire, UK

Abstract

The accurate estimation of thermal contact conductance (TCC) is a fundamental need towards the optimal design of a cryogenic cooling system for the new flagship beamline CSXID at Diamond Light Source, which utilizes mechanically pressed components of gold-plated copper to form the heat conduction path. To aid development, a study of thermally conducting joints at cryogenic temperatures has been performed combining Mathworks[®] Simulink and ANSYS Workbench[™]. To verify and validate the simulation results, an experimental setup will be made to carry out experiments to determine performance of the system, taking into consideration parameters including surface roughness, surface finish, temperature, & clamping force, which all greatly influence TCC.

BACKGROUND

The flagship beamline CSXID at Diamond Light Source is aiming to be a world leader in nano positioning and imaging resolution. The beamline will be used to perform experiments to study the magnetic and electrical properties of materials under certain environmental conditions, using the Quantum Material Imaging (QMI) endstation. Cryogenics is one requirement for this endstation, where the sample environment must be tightly controlled. In addition, the sample must rotate about two axes, have a magnetic field with an adjustable field strength and polarity, and be able to excite the sample with RF signals, all while maintaining position stability in the sub-nanometre range. Here presents a challenge; many systems with small space envelopes which must operate concurrently.

To design a cryogenic system with a high level of confidence, thermal modelling must be performed, which can also aid the development in the early design stage.

REQUIREMENTS

The system's requirements are shown in Table 1.

Table 1: Cryogenic Requirements at the Sample Position

Description	Requirement
Target sample temperature	20 K
Cooldown time after venting	3 hrs
Cooldown time after sample transfer	30 mins
Sample transfer temperature	300 K
Warm-up time	30 mins
Sample position drift during cooling	<10 μ m

[†] david.tillin@diamond.ac.uk

WORKING PROCESS

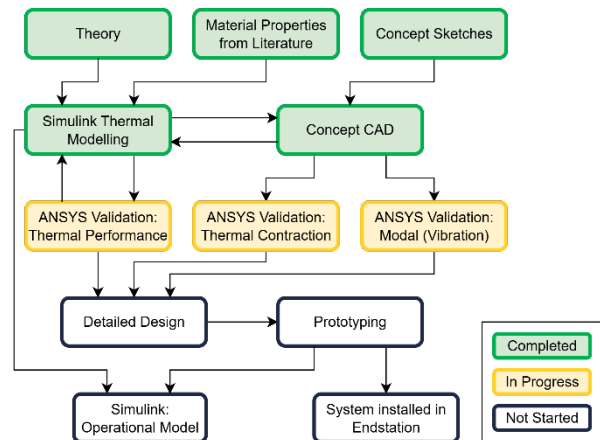


Figure 1: A flowchart showing the working process used to develop the cryogenic system.

Figure 1 shows an outline of the working process behind the development of the cryogenic system. Material properties were taken from literature to generate a model using Simulink in combination with geometrical data from the concept system. This concept is shown in Fig. 2. The Simulink model was then used to rapidly iterate the design. Simulink creates time-resolved models, which allows the prediction of cooldown times and low temperature operations.

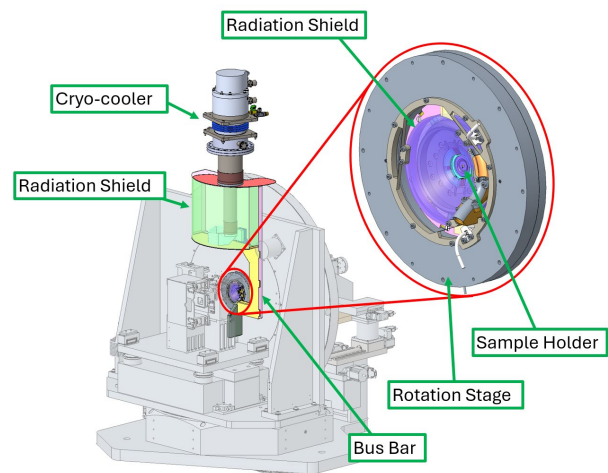


Figure 2: The QMI Endstation concept. The Cryogenic system is highlighted.

Once the design reached a satisfactory level of confidence, an ANSYS steady-state thermal analysis was used to validate the simulated results, to analyse the effects of thermal contraction on positioning, and modal analysis was used to generate expected resonant mode shapes.

The parameters assumed in both ANSYS and Simulink models will be validated by prototype testing and final commissioning. Alongside this, an “operational” Simulink model will be created, which will take the preliminary model and scale the results according to real-world test data. The model could then be used to predict the time-scales for user experiments on the commissioned beamline.

THEORY

Thermal contact conductance (TCC) quantifies how easily heat flows across the interface between two solid bodies in direct contact. TCC is difficult to accurately calculate because there are many material properties and variables involved such as thermal conductivity, joint force, surface area, surface roughness, yield strength, hardness, Poisson's ratio, and elastic modulus.

The material surfaces of a conforming rough solid-solid contact are considered. “Conforming” means the two materials share equally distributed contact area across the entire surface. “Rough” refers to the profile of the surfaces at a microscopic level, which consists of defects called asperities. The surfaces only contact at the peaks of these asperities, which deform under load. To derive the function between true contact area and surface roughness, the interface is simplified by modelling one surface as smooth and the other as rough. In this model, the asperity gradient and distribution are estimated from the combined roughnesses of both original surfaces [1].

Using these assumptions, a set of correlation equations exist for “elasto-plastic” models, where a “plasticity index” is used to determine if plastic or elastic deformation occurs, respective to the size and spacing of the asperities [2]:

$$\psi = \frac{E'}{10H_{micro}} \sqrt{\frac{\sigma}{r}}, \quad (1)$$

where E' is the equivalent elastic modulus, H_{micro} is the surface microhardness of the softer material, σ is the root mean square (RMS) surface roughness of the softer material, and r is the average contact spot radius of the deformed asperities.

The correlation equations used contain two constant coefficients. The equations were expressed for thermal contact resistance (TCR), which is simply the inverse of TCC, due to Simulink using thermal resistance elements. For the elastic case, the correlation equation is:

$$R_C = \frac{1}{A} \left(\frac{\sigma_s}{m_s} \right) \frac{1}{k_s} \left(\frac{\sqrt{2} p_{applied}}{E' m_s} \right)^{-B}. \quad (2)$$

For the plastic case, the equation is:

$$R_C = \frac{1}{A} \left(\frac{\sigma_s}{m_s} \right) \frac{1}{k_s} \left(\frac{p_{applied}}{H_{micro}} \right)^{-B}, \quad (3)$$

where A and B are the correlation coefficients, R_C is thermal contact resistance, σ_s and m_s are calculated from the RMS surface roughnesses, E' is the equivalent elastic modulus, k_s is the equivalent thermal conductivity at a given temperature, and $p_{applied}$ is the pressure of the joint based on the apparent surface area [3].

Data for surface microhardness is not available for many materials, so it has been approximated. It can be expressed in terms of yield strength and strain hardening exponent [4]. When the strain hardening exponent is set to 0, it gives a worst-case for the microhardness, thus conductance. Therefore, a simple approximation can be obtained based on yield strength alone:

$$H_{micro} \cong 3R_e, \quad (4)$$

where R_e is the yield strength of the softer material. This has been correlated against experimental data for some hard metals, where the microhardness has a greater effect on conductance than soft materials like copper [5].

THERMAL MODELLING

The model can be considered at a system level. The components are represented in Fig. 3, to capture their interactions through radiation and conduction paths. This flow chart is used as a basis for a thermal lump-mass network, which uses thermal masses to represent the components in the system, with resistances between those masses. Simulink was used to express components and their interactions using the SimScape library, which contains elements such as thermal mass, conduction, radiation, and variable resistance.

Typically, liquid helium cooling is used when there are tight stability specifications, but these are space intensive and expensive to operate. A two-stage closed-cycle cryocooler was chosen for this application (ARS Cryo GMX-20B DE210SF), due to the vibration requirements and

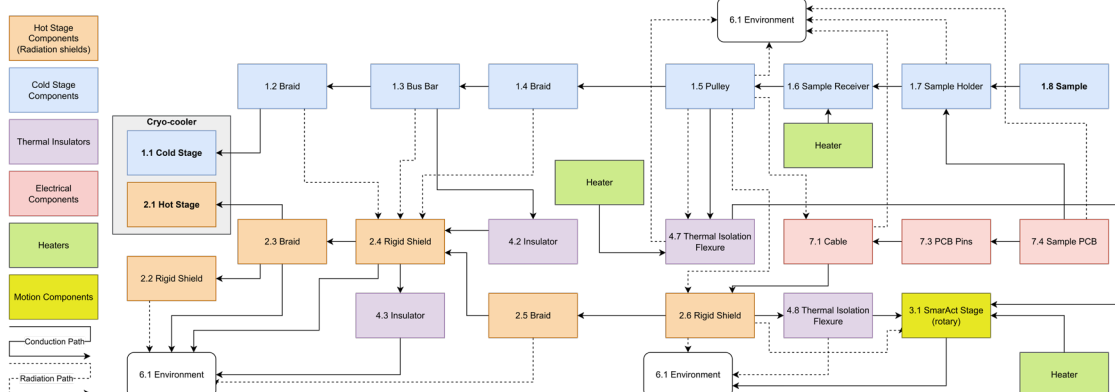


Figure 3: A flowchart depicting the cryogenic system and its interactions between components.

cooling power. This cooler mechanically isolates whilst providing a thermal conduction via a small He filled interface. To accurately simulate the cooler's temperature dependent performance, a subsystem was created with two feedback loops, using lookup tables to alter the cooling power based on the cooler's temperature.

The TCR approximation equations were used to create lookup tables for TCR vs temperature for each solid-solid interface in the system. The Simulink network measures the temperature at a given interface and outputs the thermal resistance of that interface.

The Simulink results allow the design of complex components to thermally isolate the low-temperature components. For instance, the sample holder is supported by a tripod-shaped flexure, which has a low thermal conductance due to the small cross-sectional area of the flexures. This effect is shown in Fig. 4, where the temperature gradient over the flexure can be seen isolating the cryogenic sample holder (blue) from the environment (red).

RESULTS

The Simulink model results showed that the target temperature range was achievable, with a predicted base temperature of **17.6 K**. The ANSYS simulation gave good confidence in the lump-mass model, giving a sample temperature of **18.3 K** shown in Fig. 4. The nodes measured across the thermal isolation flexures also show a similar temperature trend to Simulink.

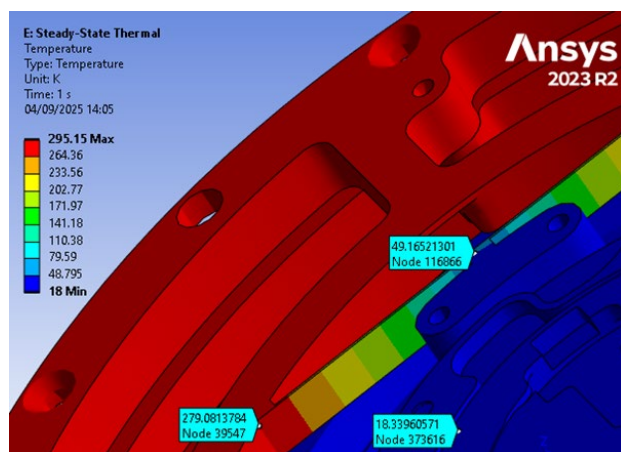


Figure 4: ANSYS thermal analysis of a thermal isolation flexure near the sample.

Cooldown times do not yet meet the requirements; the current design achieves **~7.5 hours** from ambient temperature to 50 K, shown in Fig. 5. Once the system has stabilized, the sample location is heated locally to 300 K to allow ambient temperature sample transfer, while the cryocooler is kept running. The system is then cooled again. The time taken in this scenario was **78 minutes** from 300 K to 50 K. However, this can be treated as a lower bound because the specific heat of each material is constant in Simulink where this would decrease in reality. Warmup times are easily met with the assumed 20 W heaters raising the sample temperature to 300 K in **18 minutes**.

The thermal contraction analysis showed that the sample moved by **~4 μm** between 300 and 20 K. This motion was only along the beam direction, which would manifest as negligible defocusing of the image. The modal analysis showed a first mode frequency at **407 Hz**; however, the mode shape was a rotation around the sample, which should not impact the position stability.

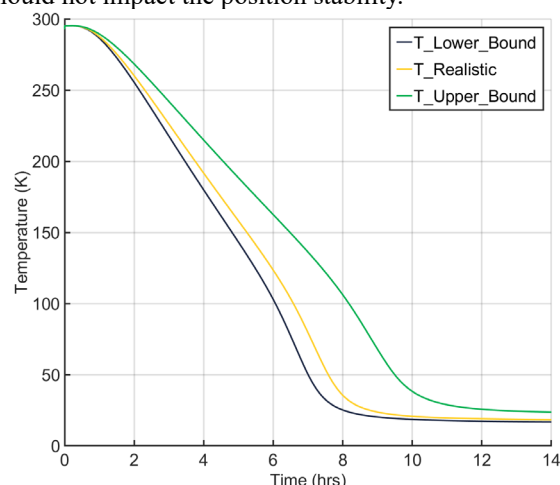


Figure 5: A plot of sample temperature vs time during the initial system cooldown from ambient temperature.

CONCLUSION

The thermal performance requirements were all met with good agreement between Simulink and ANSYS, with the exception of cooldown times. It was agreed that prototyping could begin to increase the confidence level of these results. The detailed design phase will focus on reducing cooldown times by eliminating unnecessary mass. The prototyping phase will help to optimise the design and improve cooldown times, for which a detailed testing plan has been created. Once the prototyping phase is complete, the final system can be manufactured and tested.

REFERENCES

- [1] M. M. Yovanovich, "Four decades of research on thermal contact, gap, and joint resistance in microelectronics," *IEEE Trans. Compon. Packag. Manuf. Technol.*, vol. 28, no. 2, pp. 182-206, 2005. doi:10.1109/TCAPT.2005.848483
- [2] J. A. Greenwood and J. P. Williamson, "Contact of nominally flat surfaces," *Proc. Royal Society of London*, vol. 295, no. 1442, pp. 300-319, 1966. doi.org/10.1098/rspa.1966.0242
- [3] R. C. Dhuley, "Thermal Contact Resistance," Fermi Research Alliance, Hartford CT, USA, 2019. <https://lss.fnal.gov/archive/2019/slides/fermilab-slides-19-042-td.pdf>
- [4] R. Cahoon, "An improved equation relating hardness to ultimate strength," *Metall. Trans.*, vol. 3, p. 3040, 1972. doi:10.1007/BF02652880
- [5] J. T. Busby, M. C. Hash and G. S. Was, "The relationship between hardness and yield stress in irradiated austenitic and ferritic steels," *J. Nucl. Mater.*, vol. 336, nos. 2-3, pp. 267-278, 2004. doi:10.1016/j.jnucmat.2004.09.024